HE2003 Econometrics I: Linear Regression with One Regressor 1

Wenjie Wang

Economics, School of Social Sciences
Nanyang Technological University

Aug 19, 2024

Linear Regression with One Regressor

The Linear Regression Model

Example: Class Size vs Student Performance. If we cut the class size, what will the effect be on student performance?

That is to say, if we change the class size (e.g., student-teacher ratio, STR) by a certain amount, what would be the corresponding change in test score? We can write this relationship as follows:

$$\text{Test Score} = \beta_0 + \beta_1 STR + \text{other factors} \tag{1}$$

$$\begin{aligned}\beta_1 &= \frac{\Delta \text{Test Score}}{\Delta STR} \\ &= \text{change in test score for a unit change in STR} \\ &= \text{slope of population regression line}\end{aligned}$$

We expect that β_1 is negative, but we need to obtain its precise value for policy making. Suppose that $\beta_1 = -2$, then a reduction in STR of 2 would yield an increase in test score of $(-2) \times (-2) = 4$.

Linear Regression with One Regressor

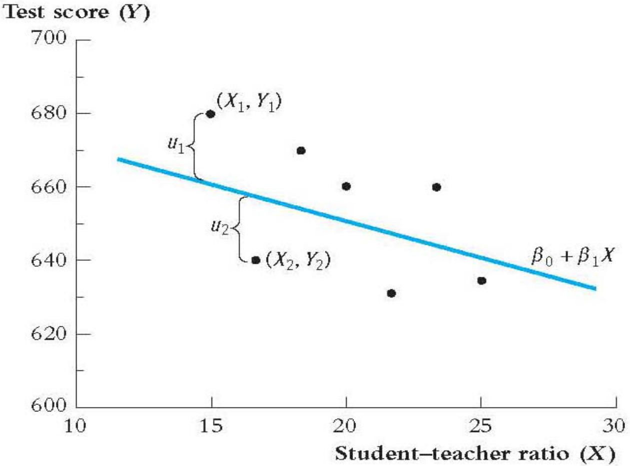
KEY CONCEPT In general, the (simple) linear regression model we consider can be written as:

$$Y_i = \beta_0 + \beta_1 X_i + u_i, \quad i = 1, 2, 3, \ldots, n,$$

- We have n observations: $(X_i, Y_i), i = 1, 2, 3, ..., n$;
- Y_i is the dependent variable or explained variable (Test score in the example);
- X_i is the independent variable or explanatory variable or regressor (STR in the example);
- $\beta_0 + \beta_1 X$ is the population regression line;
- β_0 is the intercept and β_1 the slope of the populatoin regression line;
- u_i is the error term of the regression (other factors that also influence Y).

Linear Regression with One Regressor

Illustration: Observations on Y and X (sample size n=7); the population regression line; and the error terms. u_1 and u_2 are error terms: (X_1, Y_1) has positive error u_1 and (X_2, Y_2) has negative error u_2.



Linear Regression with One Regressor

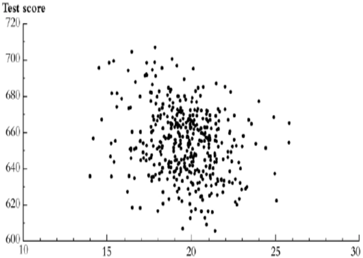
About the meaning of the intercept and the error term

- The intercept β_0 is the value of the population regression line when $X = 0$, as it is the point at which the population regression line intersects the Y axis.
- In some economic applications, β_0 has a meaningful economic interpretation. In other applications, it does not have real-world meaning. For example, when X is the class size (STR), β_0 is the predicted value of test scores when there are no students in the class (STR=0) ! In such applications, we are typically more interested in the value of β_1.
- The error term u_i incorporates all the factors responsible for the difference between Y_i and the value predicted by the population regression line ($\beta_0 + \beta_1 X_i$). For the current example with class size and test score, u_i may be the (overall) family background in district i, the (overall) quality of school facilities in district i, etc.

Linear Regression with One Regressor

Estimating β_0 and β_1 in the linear regression model

- In practice, β_0, β_1, and the population regression line are unknown. Therefore, we must use data to estimate these unknown values.
- A scatterplot of 420 observations in the California School District Data is shown below. One could somehow "eyeball" a straight line through these data. However, different people will create different lines... Then, how to find the best line? That is, how to choose the best values of β_0 and β_1? We can find such values by solving a minimization problem.



Linear Regression with One Regressor

KEY CONCEPT: The Ordinary Least Squares (OLS) Estimator chooses the values of b_0 and b_1 so that the estimated regression line is as close as possible to the observed data, where closeness is measured by the sum of squared errors (mistakes) in predicting Y given X.

- Here, given some values of b_0 and b_1, the error (mistake) in predicting Y_i given X_i can be written as: $Y_i - (b_0 + b_1 X_i)$, for each i (i from 1 to n).
- Therefore, the sum of squared errors can be written as:

$$\sum_{i=1}^{n} [Y_i - (b_0 + b_1 X_i)]^2 = [Y_1 - (b_0 + b_1 X_1)]^2 + \ldots + [Y_n - (b_0 + b_1 X_n)]^2.$$

Linear Regression with One Regressor

The OLS estimators of the true values β_0 and β_1 are obtained by solving the following minimization problem:

$$\min_{b_0, b_1} \sum_{i=1}^{n} [Y_i - (b_0 + b_1 X_i)]^2$$

- Intuitively, we search over all possible values of b_0 and b_1, and finds the values that minimize the sum of squared errors, in practice we solve the first order conditions of the minimization problem.

Linear Regression with One Regressor

More precisely, we first take the partial derivatives with respect to b_0 and b_1:

$$\begin{aligned}
\frac{\partial}{\partial b_0}\sum_{i=1}^{n}(Y_i - b_0 - b_1 X_i)^2 &= -2\sum_{i=1}^{n}(Y_i - b_0 - b_1 X_i), \qquad (2)\\
\frac{\partial}{\partial b_1}\sum_{i=1}^{n}(Y_i - b_0 - b_1 X_i)^2 &= -2\sum_{i=1}^{n}(Y_i - b_0 - b_1 X_i)X_i, \qquad (3)
\end{aligned}$$

The OLS estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ are the values of b_0 and b_1 for which (2) and (3) equal zero.

Linear Regression with One Regressor

Re-arranging (2) and (3), setting them equal to zero, and dividing by n gives:

$$\bar{Y} - \hat{\beta}_0 - \hat{\beta}_1 \bar{X} = 0, \tag{4}$$

$$\frac{1}{n}\sum_{i=1}^{n} X_i Y_i - \hat{\beta}_0 \bar{X} - \hat{\beta}_1 \frac{1}{n}\sum_{i=1}^{n} X_i^2 = 0. \tag{5}$$

where $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$ and $\bar{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i$ (the sample averages).

Linear Regression with One Regressor

Solving the two equations (4) and (5), we can obtain the formula of the OLS estimators $\hat{\beta}_0$ and $\hat{\beta}_1$: Specifically, (4) implies that $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$; Replacing $\hat{\beta}_0$ in (5) by $\bar{Y} - \hat{\beta}_1 \bar{X}$, we obtain

$$\frac{1}{n} \sum_{i=1}^{n} X_i Y_i - (\bar{Y} - \hat{\beta}_1 \bar{X}) \bar{X} - \hat{\beta}_1 \frac{1}{n} \sum_{i=1}^{n} X_i^2 = 0, \tag{6}$$

which is equivalent to

$$\frac{1}{n} \sum_{i=1}^{n} X_i Y_i - \bar{X} \bar{Y} + \hat{\beta}_1 \bar{X}^2 - \hat{\beta}_1 \frac{1}{n} \sum_{i=1}^{n} X_i^2 = 0. \tag{7}$$

Then, we can solve (7) to obtain the formula for $\hat{\beta}_1$.

Linear Regression with One Regressor

KEY CONCEPT: Formula of the OLS Estimator

By solving the minimization problem, we obtain:

$$\begin{aligned} \hat{\beta}_1 &= \frac{\frac{1}{n}\sum_{i=1}^{n} X_i Y_i - \bar{X}\bar{Y}}{\frac{1}{n}\sum_{i=1}^{n} X_i^2 - \bar{X}^2} = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^{n}(X_i - \bar{X})^2} = \frac{S_{XY}}{S_X^2}, \quad (8) \\ \hat{\beta}_0 &= \bar{Y} - \hat{\beta}_1 \bar{X}, \end{aligned}$$

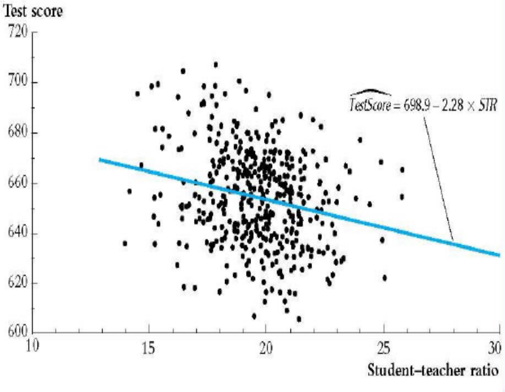
where $S_{XY} = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})$ (the sample covariance between X and Y), and $S_X^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$ (the sample variance of X).

- $\hat{\beta}_0$ is called the estimated intercept, and $\hat{\beta}_1$ the estimated slope.
- $\hat{\beta}_0$ and $\hat{\beta}_1$ are computed from a sample of n observations of X_i and Y_i, $i = 1, \ldots, n$.
- $\hat{\beta}_0$ and $\hat{\beta}_1$ are estimates of the unknown true values β_0 and β_1, respectively.

Linear Regression with One Regressor

Application to the California Test Score - Class Size Data (caschool.dta)

- Estimated slope $= \hat{\beta}_1 =$ -2.28;
- Estimated intercept $= \hat{\beta}_0 =$ 698.9;
- Estimated regression line: Test Score = 698.9 - 2.28 $\times$ STR.



Linear Regression with One Regressor

OLS regression: STATA output

```
regress testscr str, robust

Regression with robust standard errors               Number of obs =     420
                                                      F(  1,   418) =   19.26
                                                      Prob > F      =  0.0000
                                                      R-squared     =  0.0512
                                                      Root MSE      =  18.581

-------------------------------------------------------------------------
        |               Robust
testscr |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
--------+----------------------------------------------------------------
    str |  -2.279808   .5194892    -4.39   0.000    -3.300945   -1.258671
  _cons |    698.933   10.36436    67.44   0.000     678.5602    719.3057
-------------------------------------------------------------------------
```

Linear Regression with One Regressor

Intepretation of the Estimated Slope and Intercept

$$\widehat{\text{Test Score}} = 698.9 - 2.28 \times STR$$

On average, districts with one more student per teacher have test scores that are 2.28 points lower (This makes sense: Larger class size lower test scores, smaller class size higher test scores). That is,

$$\frac{\Delta \text{Test Score}}{\Delta STR} = -2.28.$$

- Now, given certain value of STR (e.g., asked by the superintendent, our client), we may predict the value of Test Score:

$$\widehat{\text{Test Score}} = \hat{\beta}_0 + \hat{\beta}_1 \times STR.$$

- For example, for a school district with 20 students per teacher ($STR = 20$), the OLS predicted test score is:

$$698.9 - 2.28 \times 20 = 653.3.$$

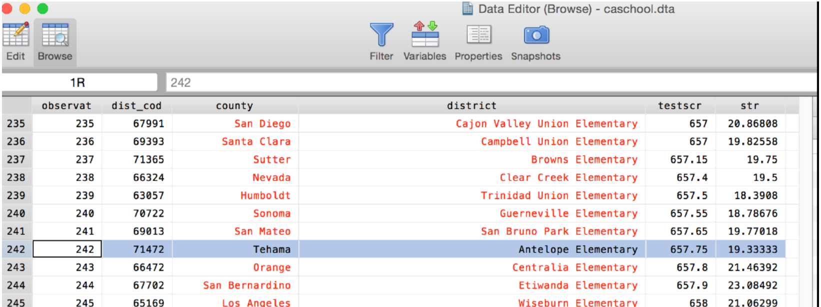
Linear Regression with One Regressor

KEY CONCEPT: OLS Predicted Values and Residuals

We can also estimate the error term u_i using OLS.
For example, one of the districts in the data set is Antelope, CA, for which $STR = 19.33$, and Test Score $= 657.8$.

- Predicted value: $\widehat{\text{Test Score}}_{Antelope} = 698.9 - 2.28 \times 19.33 = 654.83$,
- Residual: $\hat{u}_{Antelope} = \text{Test Score}_{Antelope} - \widehat{\text{Test Score}}_{Antelope} = 657.75 - 654.83 = 2.92$.

  Q: What does the positive residual of Antelope tell us?



	observat	dist_cod	county	district	testscr	str
235	235	67991	San Diego	Cajon Valley Union Elementary	657	20.86808
236	236	69393	Santa Clara	Campbell Union Elementary	657	19.82558
237	237	71365	Sutter	Browns Elementary	657.15	19.75
238	238	66324	Nevada	Clear Creek Elementary	657.4	19.5
239	239	63057	Humboldt	Trinidad Union Elementary	657.5	18.3908
240	240	70722	Sonoma	Guerneville Elementary	657.55	18.78676
241	241	69013	San Mateo	San Bruno Park Elementary	657.65	19.77018
242	242	71472	Tehama	Antelope Elementary	657.75	19.33333
243	243	66472	Orange	Centralia Elementary	657.8	21.46392
244	244	67702	San Bernardino	Etiwanda Elementary	657.9	23.08492
245	245	65169	Los Angeles	Wiseburn Elementary	658	21.06299

Linear Regression with One Regressor

- In general, the OLS predicted value (fitted value) is defined as:

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i, \;\; i = 1, ..., n.$$

  For the CA data, we can define the OLS predicted Test Score for each district:

$$\widehat{\text{Test Score}}_i = \hat{\beta}_0 + \hat{\beta}_1 STR_i, \;\; i = 1, ..., 420.$$

- The OLS residual is defined as the difference between actual value of Y_i and predicted value of Y_i:

$$\hat{u}_i = Y_i - \hat{Y}_i = Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_i), \;\; i = 1, ..., n.$$

- $\hat{u}_i$ is estimate of the unknown true error term u_i for each observation i, where $i = 1, ..., n$.